

Analysis of Streaming Movie Data Part 2

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1 Report

1.1 Reflection of data visualisation and Python in general

Data visualisation is crucial in data analysis, providing visual insights into patterns and trends during both exploratory and explanatory phases. In the exploratory phase, visualisations confirm data suitability, check for anomalies, detect outliers, and explore patterns and correlations. In the explanatory phase, visualisations aid in communicating findings to enhance decision-making and create change.

Python, an open-source language, offers visualisation capabilities through libraries like Matplotlib, Seaborn, and Bokeh. Compared to other tools, Python is cost-effective, easy to use, and has extensive community support, providing flexible and customisable visualisation tools. However, Python carries risks such as security vulnerabilities, lack of official support, and compatibility issues. Handling sensitive data requires compliance with privacy laws to avoid breaches. Users must ensure responsible, ethical use by validating tools, protecting data, and maintaining transparency in their work (Stancin, 2019, Wazarkar 2021).

Matplotlib is a two-dimensional plotting library built on NumPy arrays. It allows the creation of various plots with customisation options. The main strength is the level of control in the syntax to provide complex and detailed visualisations. However, the syntax can be cumbersome, as Matplotlib is procedural and thus each step needs to be provided. Its default visual output is also less aesthetically pleasing compared to more modern libraries like Seaborn (Sial et al, 2021).

Seaborn, built on Matplotlib, simplifies creating complex visualisations by providing a high-level interface. It is more imperative and semantic. It integrates with Pandas data structures making it easy to work with dataframes. Seaborn is excellent in exploratory data analysis, allowing automatic handling of statistical transformations with minimal syntax. However, it is less flexible for highly customised visualisations compared to Matplotlib (Sial et al, 2021, Waskom 2021).

Bokeh allows the creation of interactive dynamic visualisations that can be displayed on the web and handle real-time data updates. However, Bokeh has fewer visualisation options, requires a steeper learning curve and is still undergoing changes. It also requires importing various libraries to produce plots which can be cumbersome compared to Matplotlib and Seaborn which only requires one library to be imported (Sial et al, 2021).

Each library offers unique strengths and selecting the right one depends on the specific context and visualisation needs.

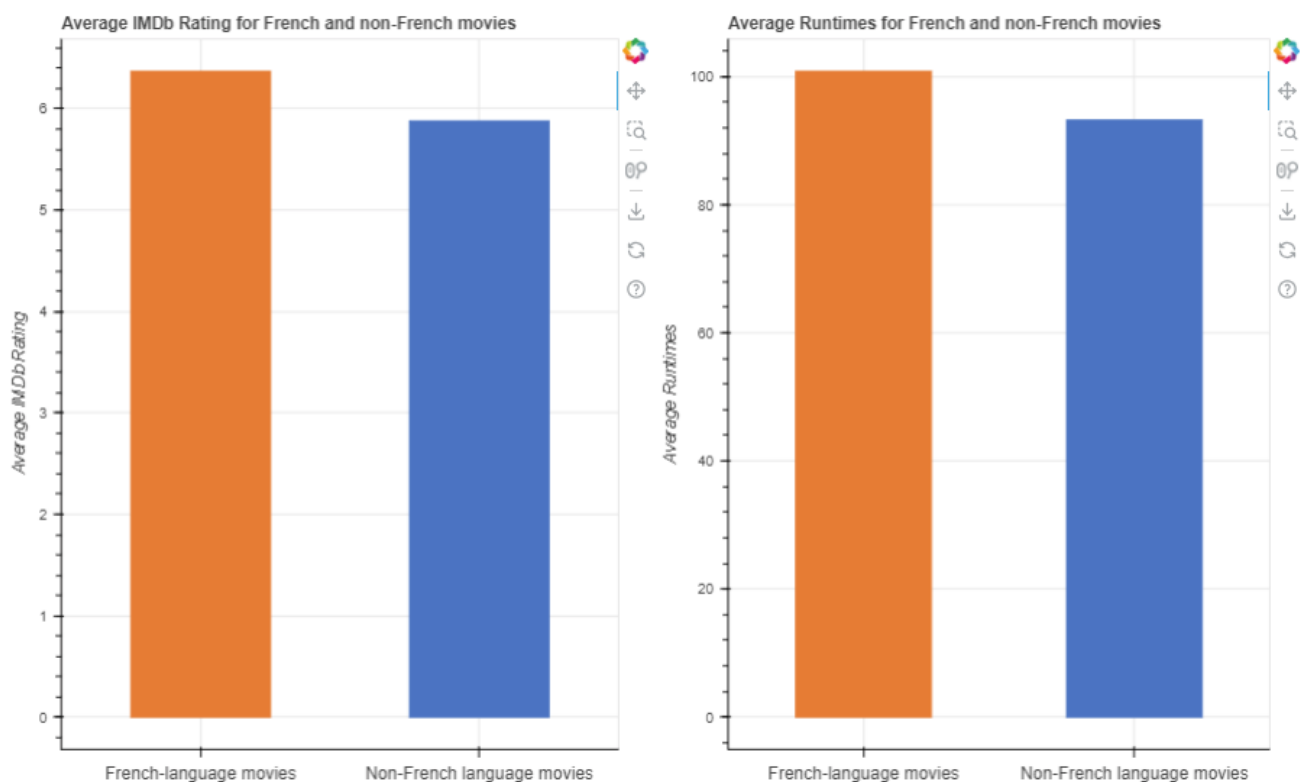
1.2 Justification of approach to create data visualisation dashboard

For A) business goal, B) exploration of data and C) assessing quality of data, please refer to Jupyter notebook

D. Performing analyses to address the business goals

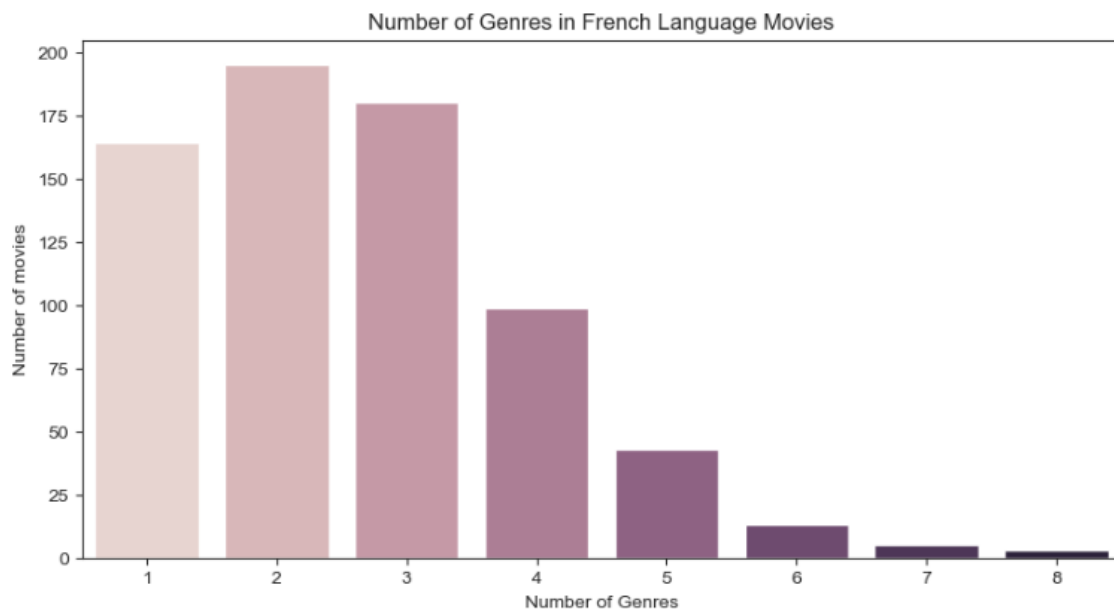
1. Does France make longer movies on average based on the runtime? & 2. Does France make better movies on average based on ratings?

Two methods were used to answer the business questions. Method 1 used the Country variable, assuming the first listed country is the primary one. Method 2 used the Language variable, filtering for French and non-French movies. For Method 1, means of runtime and IMDb ratings were plotted for each country using Matplotlib. In hindsight, Seaborn would have been better as it calculates means automatically. However, Method 2 proved more effective for these business questions. It allowed plotting average IMDb ratings and runtimes for French and non-French movies using Bokeh. However, Matplotlib/Seaborn is preferred for ease of use, as there was no need to call different libraries.



3. Are there any French movies that claim to be of more than one genre?

Two methods were used to answer this question. Method 1 used the split method to split the genres and count the number of genres per movie only. Method 2 used the explode method to split the genres into separate rows then use groupby to count number of genres per movie. Both produced the same outcome with the box plot created using Seaborn (see below). Seaborn was simple to use and provided a clear visual of the outcome.

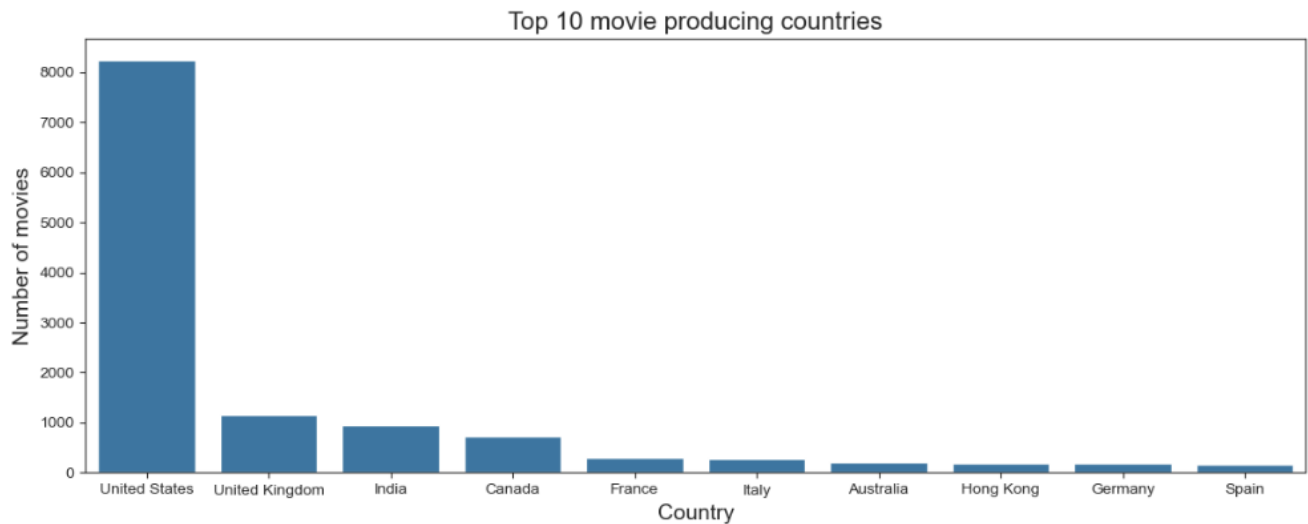


Using Method 2, one can identify the movies with more than one genre. Below are the movies with more than six genres:

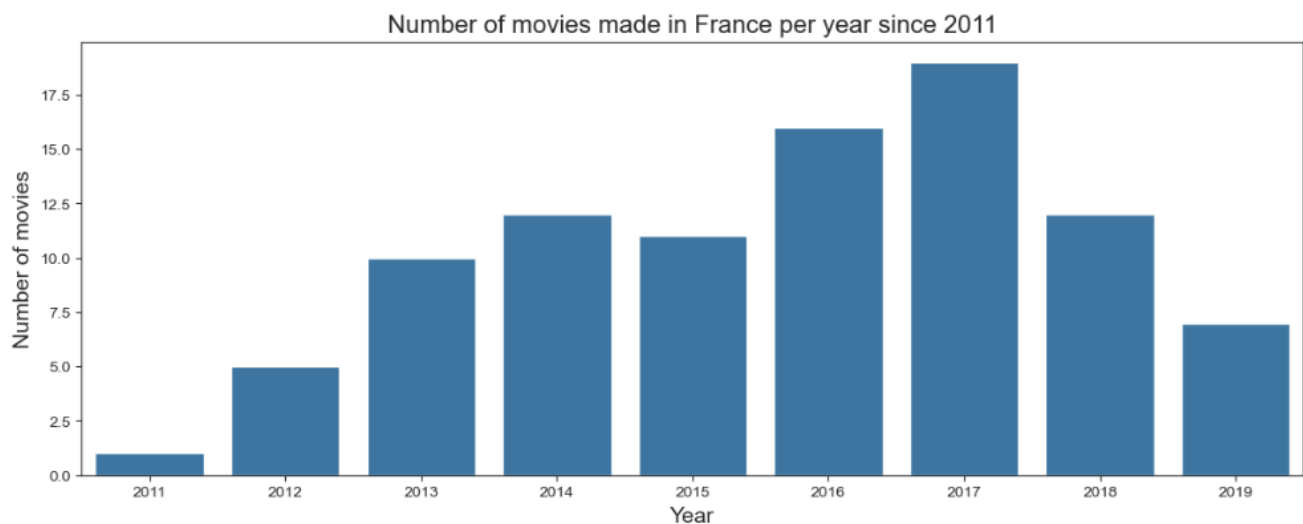
	Title	genre_number
635	The Tourist	6
602	The Pink Panther 2	6
55	Asterix the Gaul	6
59	Atlantis: The Lost Empire	6
177	Edison & Leo	6
215	Ghosts of Cité Soleil	6
503	The Ballad of Buster Scruggs	6
252	Home	6
269	Inferno	6
465	Slugterra: Ghoul from Beyond	6
464	Slow West	6
404	Phantom Boy	6
427	Ratatouille	6
513	The Brothers Grimm	7
289	Kabala	7
606	The Princess and the Frog	7
240	Herbie Goes to Monte Carlo	7
186	Ernest & Celestine	7
50	April and the Extraordinary World	8
351	Mohawk	8
43	Anastasia	8

4. Where is France placed compared to other countries with regard to the number of movies they make?

For this question, the 'new_movie_details3' dataframe was used, with the assumption that the first country listed in the 'country' column is where the movie was first made (Method 1 for questions 1 & 2). A bar plot showing the number of movies per country was created using Seaborn, which was useful for sorting values with minimal syntax. The top 10 movie-producing countries are as follow:



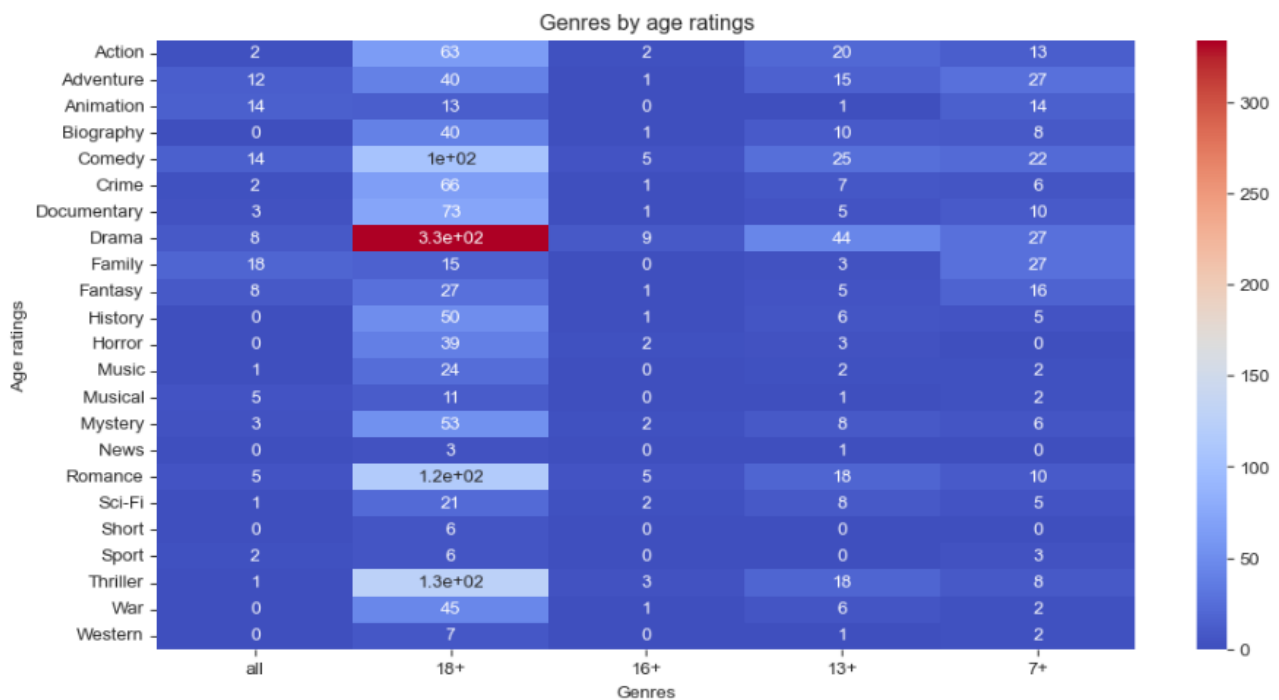
The number of movies France produced over the years was also investigated. A static bar plot was created using Seaborn, which could generate a count plot directly from the original dataframe without performing a groupby function for years:



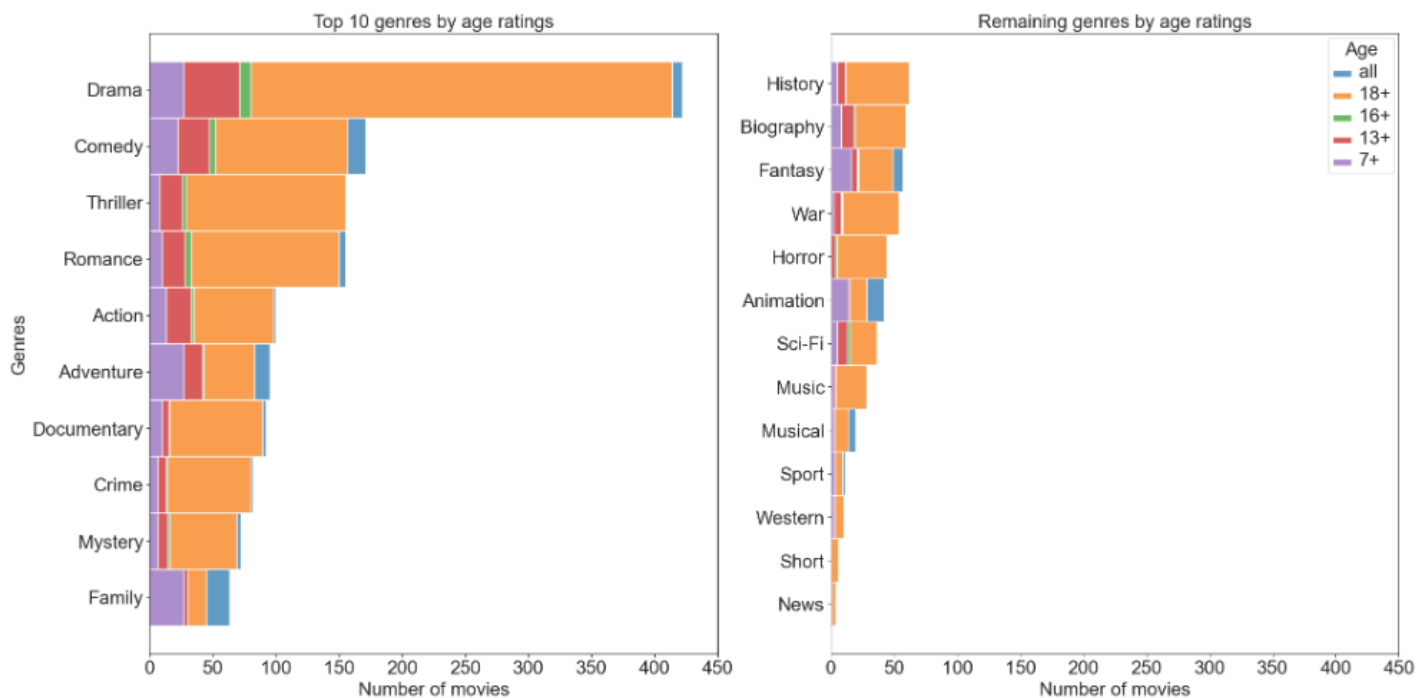
5. Is there a relationship between age groups and genres for french-language movies?

The variables of interest are Age and Genres. The 'Age' variable has 5 unique categories (All, 18+, 16+, 13+, 7+). With over 8,00 missing 'Age' values, they were replaced with 18+ to avoid inappropriate movies for younger audiences. For the 'Genre' variable, many movies have multiple genres. The exploded dataframe, which splits genres into separate rows, was used and filtered for French-language movies. Seaborn was used to explore different plots, including a heatmap and a stacked bar chart. The stacked bar chart provided a clearer visual than the heatmap:

Heatmap



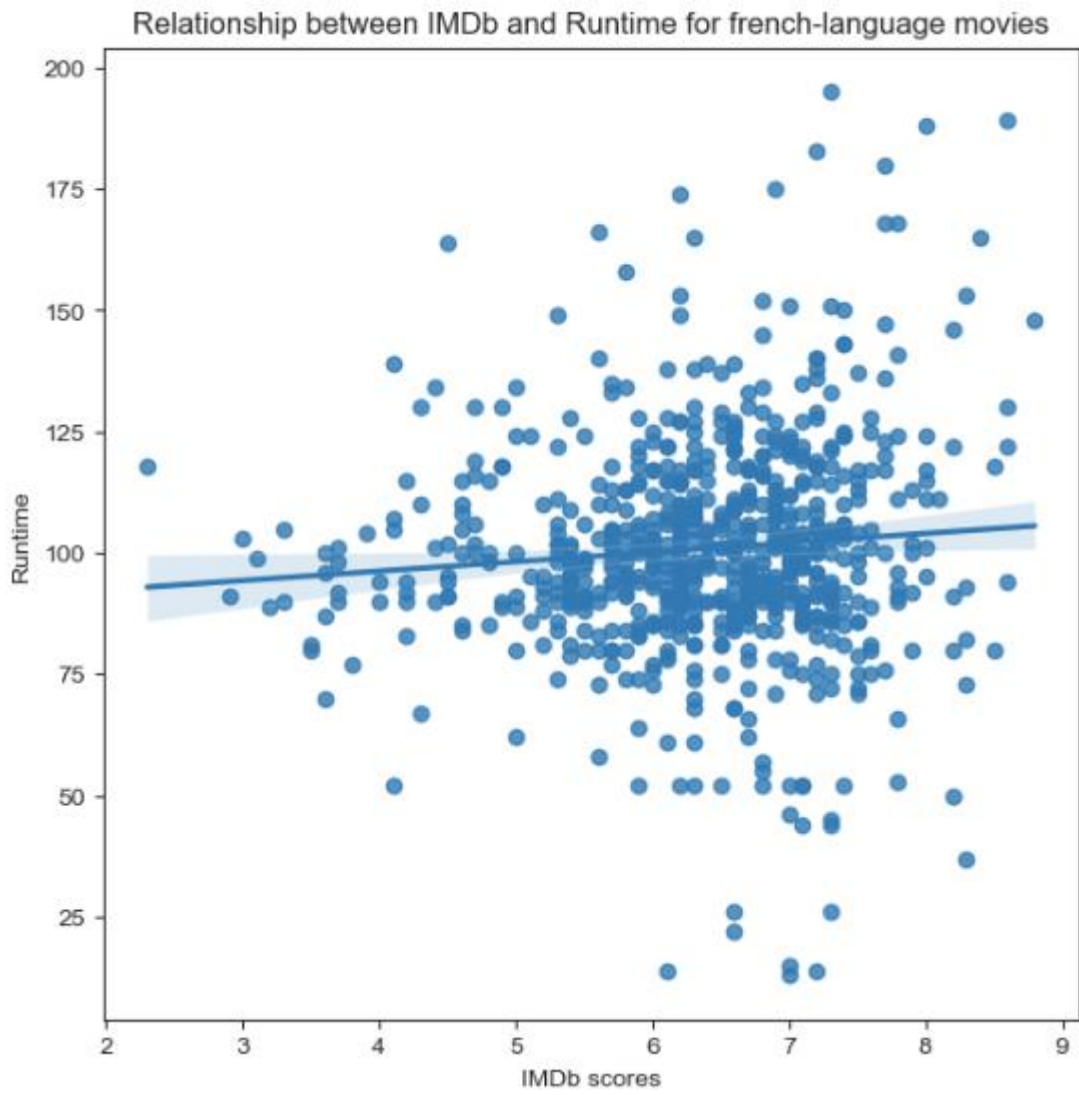
Stacked bar chart



It was ensured that the x-axis was the same scale for both figures for comparison purposes.

6. Is there a relationship between Runtime and IMDb rating for french-language movies?

A Seaborn regplot was used to create a scatter plot with a regression line for runtime and IMDb rating, segmented by age. A very weak relationship was found between IMDb and Runtime for French-language movies, indicating that longer movies tend to have higher IMDb scores.

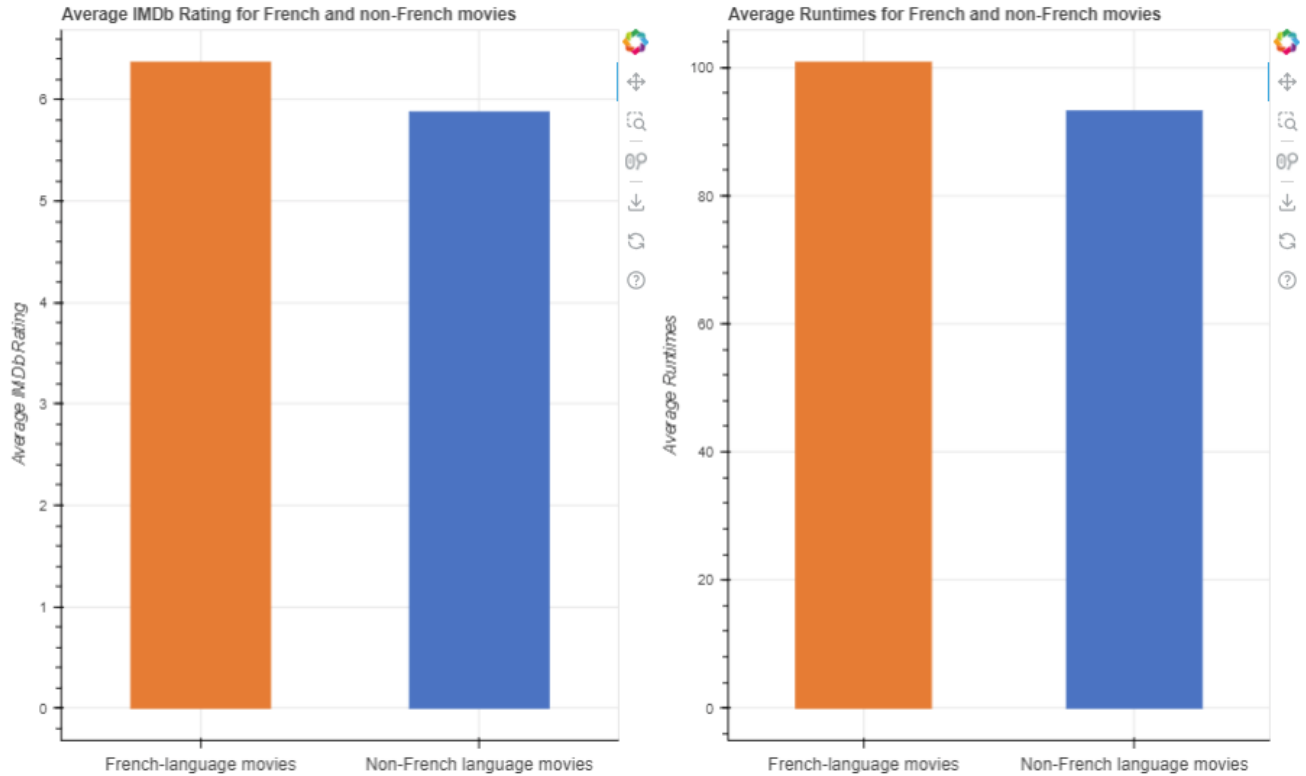


1.3 Conclusion

A dashboard was provided to Canopy with the following visuals to address their business questions.

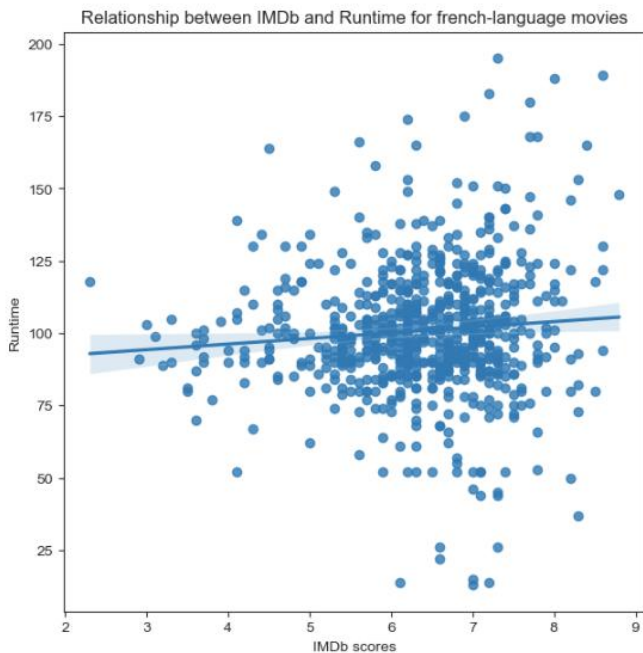
As shown in Figure 1, French-language movies had higher IMDb ratings and higher runtime than non-French language movies.

Figure 1 Does France make longer movies/better movies on average? (Question 1 & 2)



A very weak relationship was found between IMDb and Runtime for French-language movies, indicating that longer movies tend to have higher IMDb scores (Figure 2).

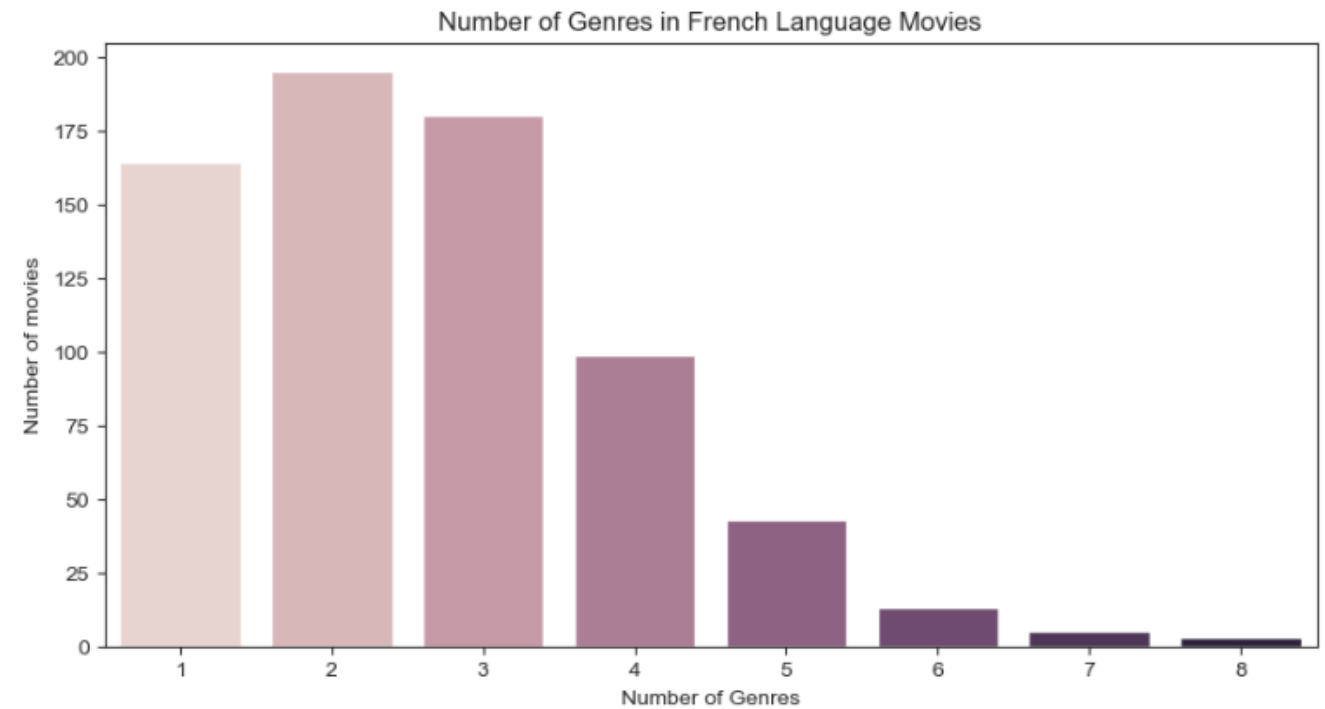
Figure 2 Is there a relationship between Runtime and IMDb rating for french-language movies? (Question 6)



As shown in

Figure 4, only 164 movies out of the 702 French-language movies had only one genre. The rest had from 2 to 8 genres.

Figure 3 Are there any French-language movies that claim to be of more than one genre? (Question 3)



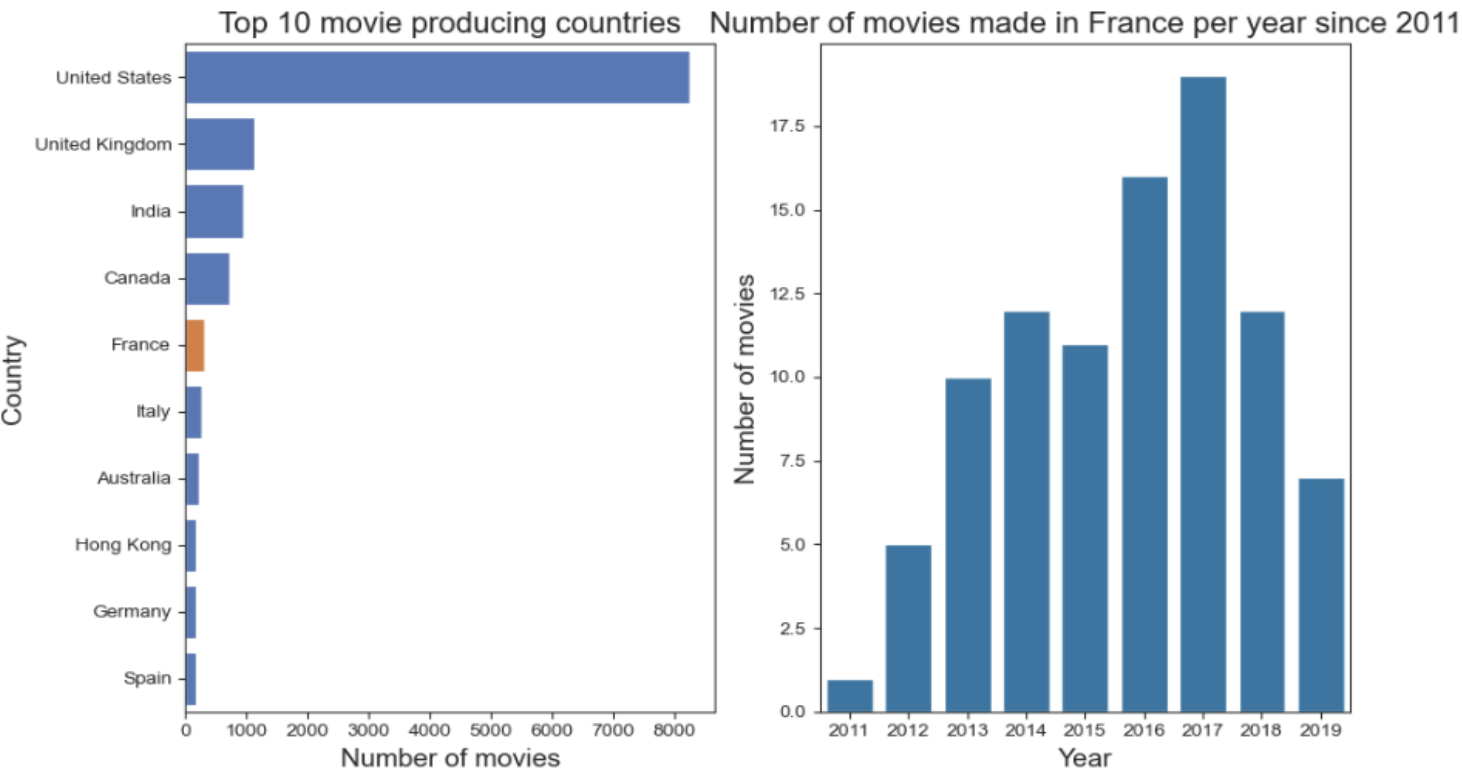
Canopy were provided the list of movies with more than six genres:

	Title	genre_number
635	The Tourist	6
602	The Pink Panther 2	6
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As shown in Figure 4, France ranks 5th in movie production, ahead of Australia. From 2011 to 2017, French movie production increased steadily but has since declined. The decline continued with the impact of COVID-19

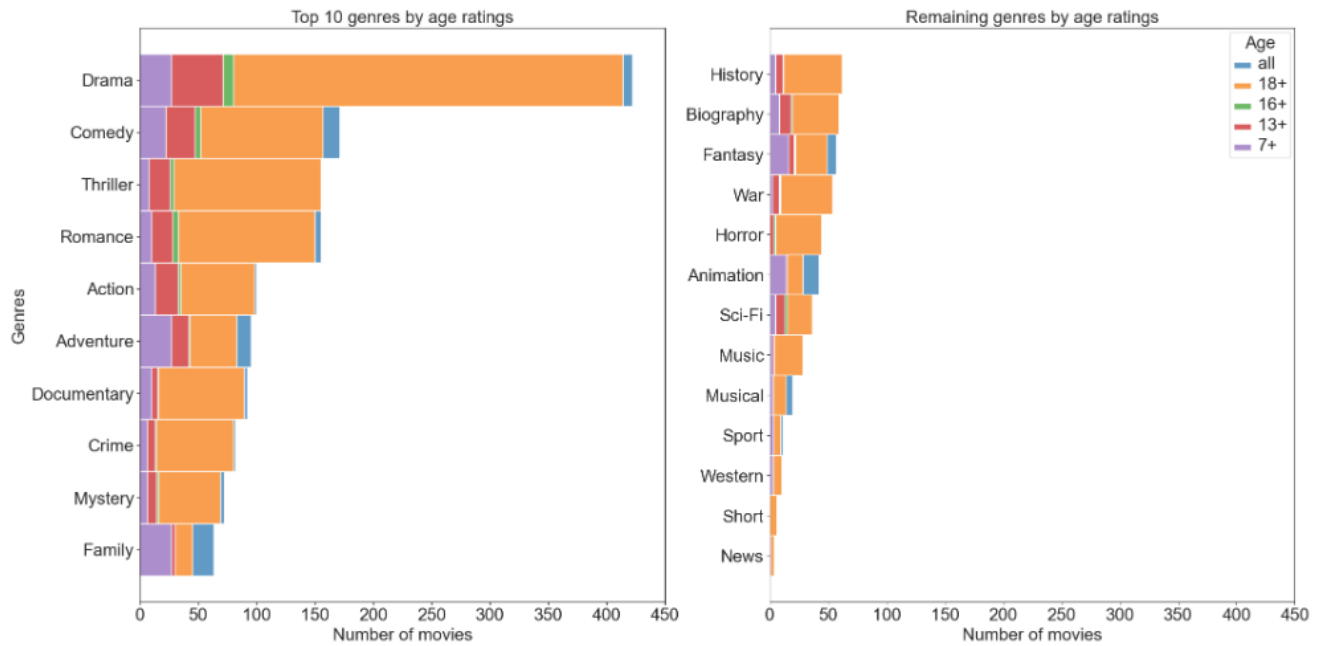
in 2020, making it unlikely that France is currently producing many movies. Therefore, Canopy should focus on producing more French films to succeed.

Figure 4 Where is France placed compared to other countries with regard to the number of movies they make? (Question 4)



Drama is the most prevalent genre across all age ratings except for the 7+ category. Comedy, Adventure, Family and Animation cater to all age groups the most. While genres like Biography, History, Fantasy, War, Animation, Horror, Sci-Fi, Musical, Sport, Western, and Shorts have fewer movies, they cater to more specific audiences. Additionally, there are fewer movies made for the 16+ age group. As the age rating increases (from 7+ to 18+), the variety of movie genres also increases, suggesting that older audiences have a wider range of genres to choose from compared to younger audiences (refer to Figure 5).

Figure 5 Is there a relationship between age groups and genres for french-language (Question 5)



Canopy appears well-positioned to achieve its business goals. The analysis indicates that French-language movies generally have higher IMDb ratings and longer runtimes compared to non-French language movies, which supports Canopy's goal of offering curated, highly rated content. Additionally, the diverse range of genres across different age groups, allows Canopy to cater to various demographics effectively. Insights into less tapped genres and the relationship between runtime and IMDb ratings help filmmakers create original content tailored for Canopy. Filmmakers should focus on genres like Biography, History, Fantasy, War, Animation, Horror, Sci-Fi, Musical, Sport, Western, and Shorts and target the 16+ age group. The decline in French movie production suggests that Canopy should produce more French films to maintain content flow. Canopy can also leverage the scarcity of new French films to position itself as a unique platform that not only curates the best existing content but also supports the creation of new, high-quality French-language movies.

2 References

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